

Poisson

The Poisson distribution density is defined as follows:

$$p(x) = \frac{\lambda^x \exp(-\lambda)}{x!}$$

...where $x = 0, 1, 2, \dots$. The mean and variance is λ .

```
> d <- rpois(10, 0.9)
> d
[1] 2 2 1 3 1 2 1 1 0 0
```

The `dpois()` function gives the log density while the `ppois()` function gives the log distribution function. A couple of examples are given below:

```
> dpois(d, 0.9)
[1] 0.16466071 0.16466071 0.36591269 0.04939821 0.36591269
0.16466071
[7] 0.36591269 0.36591269 0.40656966 0.40656966

> ppois(d, 0.9)
[1] 0.9371431 0.9371431 0.7724824 0.9865413 0.7724824
0.9371431 0.7724824
[8] 0.7724824 0.4065697 0.4065697
```

The Poisson distribution function specific parameter is λ , as shown below.

Arguments	Description
<code>lambda</code>	A vector of non-negative means

The Poisson CDF can be plotted in a graph, as shown below:

```
> x <- seq(-0.5, 5, 0.05)

> head(x)
[1] -0.50 -0.45 -0.40 -0.35 -0.30 -0.25

> plot(x, ppois(x, 1), type = "s", ylab = "F(x)", main =
"Poisson CDF")
```

Weibull

The shape parameter ' a ' and scale parameter ' b ' define the Weibull distribution density, as follows:

$$f(x) = \frac{a}{b} \left(\frac{x}{b}\right)^{a-1} \exp\left(-\left(\frac{x}{b}\right)^a\right)$$

...where $x > 0$. The `dweibull()` function gives the density while the `rweibull()` function generates random deviates, as illustrated below:

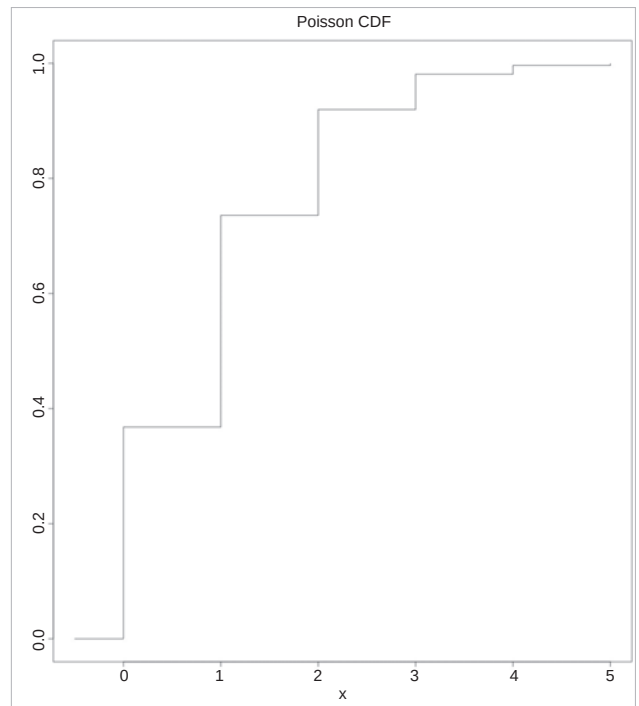


Figure 2: `ppois` plot

```
> x <- c(0, rlnorm(10))
> x
[1] 0.0000000 0.5194988 2.2301132 0.1844523 0.5267321
0.5487738 1.3437871
[8] 2.0694222 3.6199835 6.3532987 5.9563711
```

```
> all.equal(dweibull(x, shape=1), dexp(x))
[1] TRUE
```

```
> rweibull(10, 5, 1)
[1] 0.5197291 0.6851027 1.0279023 1.1397738 0.8810083
0.7945575 0.5939060
[8] 0.8762381 0.7979457 0.6977511
```

The specific parameters for the Weibull distribution are as follows.

Arguments	Description
<code>shape, scale</code>	shape and scale parameters

You are encouraged to read the manual pages for the above R functions to learn more on their arguments, options and usage. 

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